FIT5120 Industry Experience Studio Project

Data Management Plan

Team T21

Contents :

1. Datasets Overview **4**
1.1 Data Sources : 4
1.1.1 Research Papers 4
1.2 Open Data Sources 5
1.2.1 Video Game Sales Dataset 5
1.2.2 Online Gaming Behavior Dataset 5
1.2.3 Research Paper: What Makes Male Gamers Angry, Sad, Amused, and Enthusiastic While Playing Video Games? 6
1.2.4 Research Paper: The Emotional Impact of Game Duration: A Framework for Understanding Player Emotions in Extended Gameplay Sessions 6
1.3 – User-Generated Data 7
2. Iteration 3 **8**
2.1 Data Usage 8
2.1.1 Video Game Sales Dataset (DataCamp) 8
2.1.2 Online Gaming Behavior Dataset (Kaggle) 8
2.1.3 Research Paper 1 – Game Genre and Emotional Responses (ScienceDirect) 8
2.1.4 Research Paper 2 – Emotional Impact of Gameplay Duration (arXiv) 9
2.1.5 Steam Dataset – Game Metadata for Content Classification 9
2.1.6 RAWG Video Games API – Metadata Expansion & Machine Learning Input 9
2.1.7 User Input Logs 10
2.2 Data Preparation 10
2.2.1 Video Game Sales Dataset (DataCamp) 10
2.2.2 Online Gaming Behavior Dataset (Kaggle) 11
2.2.3 Research Paper 1 – Game Genre and Emotional Responses 11
2.2.4 Research Paper 2 – Emotional Impact of Gameplay Duration 11
2.2.5 Steam Dataset – Data Cleaning and Preprocessing: 12
Libraries Used: pandas, re, datetime 12
2.2.6 RAWG API – Dynamic Data Integration 12
2.3 Data Storage 13
User Input Logs Table 14
2.4 Data Archiving & Versioning 15
2.4.1 Folder Structure and Organization 15
2.4.2 Archiving Tools 16
2.4.3 Backup & Redundancy 16
2.5 Database Design 16
2.5.1 Key Entities: 16
2.5.2 Tools & Implementation: 17
2.6 Data Analytics and Modelling Techniques 18
2.6.1 Hindsight Insights (Historical Patterns) 18

2.6.2 Foresight Insights (Predictive Feedback) 18
2.6.3 Visual Insights (Dashboard & Charts) 19
2.6.4 Critical Thinking Behind Modeling Choices 19
3. Ethical, Legal, and Privacy Considerations 20
3.1 Privacy and Data Handling 20
3.2 Legal Considerations 21
3.3 Ethical Use of AI & Algorithmic Bias 22
3.4 Ethical Intent of the Project 22
4. Understanding the Problem Domain & Future Challenges 23
4.1 Problem Domain: Online Gaming, Emotional Wellbeing, and Adolescent Behavior 23
4.2 User Needs Addressed 23
4.3 Future Integration & Challenges 24
4.3.1 Planned Dataset Additions 24
4.3.2 Anticipated Challenges and Strategies 24

1. Datasets Overview

To effectively design a playful, scenario-based platform that teaches empathy and respectful online behavior, this project leverages both national-level child well-being statistics and real-world examples of online communication. The datasets selected offer insights into the prevalence, context, and patterns of bullying—particularly cyberbullying—among school-aged children in Australia.

1.1 Data Sources :

Name	Access Format	Source Update Frequency	System Update Frequency	Granularity	License/Use Conditions
Video Games Sales Data	XLSX	Static	Manual as needed	High (game level)	Creative Commons BY 4.0
Predict Online Behaviour Dataset	CSV	Static	As needed	High (session level)	Creative Commons BY 4.0
Steam Store Games	CSV	Static	As needed	High (game-level detail)	Creative Commons BY 4.0

1.1.1 Research Papers

Two research papers were used to generate rule-based datasets:

- **Research Paper 1 (ScienceDirect)** → Genre-to-emotion mappings, saved as emotion_rules_genre.csv.
- **Research Paper 2 (arXiv)** → Playtime-intensity thresholds, saved as duration_rules.csv.

These rule tables were added to the Emotion_Prediction table in the database to power emotion predictions alongside the primary datasets.

Name	Access Format	Source Update Frequency	System Update Frequency	Granularity	License/Use Conditions
Research Paper 1 (Genre-Emotion Mapping)	PDF (processed into CSV)	Static	As needed (manual)	Genre-to-emotion rules	Licensed article (ScienceDirect); manual extraction, educational use
Research Paper 2 (Duration-Intensity Mapping)	PDF (processed into CSV)	Static	As needed (manual)	Playtime-intensity rules	Open-access (arXiv); manual extraction, educational use

1.2 Open Data Sources

1.2.1 Video Game Sales Dataset

- **Description**: Includes metadata on thousands of video games including genre, platform, publisher, and global sales.
  Link: Video Game Sales – DataCamp

- **Key Variables**:

1. **Name** → Game title
2. **Platform** → System the game runs on (e.g., PS4, PC)
3. **Genre** → Used for emotion mapping (e.g., FPS, Puzzle)
4. **Publisher** → Context for parental filters or reporting
5. **Global_Sales** → Popularity indicator (optional for recommendations)
6. **Year_of_Release** → May help contextualize older vs newer games

Use in Project:

- Provides accurate, structured information about game genres.
- Helps categorize the emotional impact of different types of games.
- Offers real-world game references for generating session logs and emotion trends.

1.2.2 Online Gaming Behavior Dataset

Description: Simulated or logged user play sessions showing how teens engage with different games.

Link:
https://www.kaggle.com/datasets/rabieelkharoua/predict-online-gaming-behavior-dataset

Key Variables:

1. **Player_ID** – Unique identifier for each player
2. **Age** – Age of the player
3. **Gender** – Gender of the player (Male/Female/Other)
4. **Game_Genre** – Genre of the game
5. **PlayTime_Hours** – Total hours spent in the session
6. **Session_Per_Week** – Total sessions per week
7. **Engagement_Level** – Categorical variable (e.g., Low, Medium, High)

Use in Project:

- Tracks per-session data to evaluate how long children play and what type of games they engage with.
- Acts as the core behavioral log that is used to trigger emotion prediction.

1.2.3 Research Paper: What Makes Male Gamers Angry, Sad, Amused, and Enthusiastic While Playing Video Games?

Format: Peer-reviewed journal article
Source: ScienceDirect

Description:

- Analyses the emotional states (e.g., anger, sadness, joy) triggered by various game genres.
- Focuses on behavioural psychology and the emotional responses of gamers during different game types.

Use in Project:

- Establishes genre-based emotional mappings.
- Supports emotion classification logic for sessions based on the type of game played.
- Serves as a scientific justification for predictions shown to parents.

- Supports time-series visualizations for the parent dashboard.

1.2.4 Research Paper: The Emotional Impact of Game Duration: A Framework for Understanding Player Emotions in Extended Gameplay Sessions

Format: Preprint article (open-access)
Source: arXiv.org

Description:

- Investigates how increasing playtime duration affects emotional intensity.
- Highlights that emotions become more extreme as gameplay length increases, especially beyond 90–120 minutes.

Use in Project:

- Used to define playtime thresholds for emotion intensity scores.
- Informs backend logic for predicting emotional escalation due to overplay.
- Justifies the trigger points for parental alerts and visual cues in the dashboard.

1.2.5 Steam Dataset

This data contains detailed metadata on over 27,000 PC games available on the Steam platform. It includes information such as game titles, release dates, developers, supported platforms, age ratings, pricing, categories, genres, user ratings, and tags.

It offers a comprehensive snapshot of the digital game marketplace and is especially valuable for content analysis, violence classification, pricing trends, and age-appropriate filtering.

Link: Steam Store Games Dataset (Kaggle)

Key Variables:

- **name** – Game title
- **release_date** – Date the game was made available
- **platforms** – OS compatibility (Windows/Mac/Linux)
- **required_age** – Minimum age to play
- **categories, genres, steamspy_tags** – Gameplay descriptors and thematic tags
- **price** – Current Steam price (USD)
- **owners** – Estimated ownership range (e.g., “1,000,000 – 2,000,000”)
- achievements, positive_ratings, negative_ratings, average_playtime

Use in Project:

- Enables detailed analysis of gaming trends and metadata filtering.
- Allows for rule-based classification of games based on content severity (e.g., violence level).
- Complements emotional prediction logic by offering a structured way to evaluate game intensity and appropriateness.

1.3 User-Generated Data

The platform collects structured, voluntary session logs from authenticated users (parents) through a multi-step modal interface. These include:

- Self-reported **emotion/mood** post-gameplay
- **Game name** and **genre**
- **Date of gameplay**
- **Duration** in hours and minutes

This data supports emotion trend analysis, and personalized visual feedback. All input is stored securely and linked to a pseudonymized user identifier.

2. Iteration 3

2.1 Data Usage

In this project, data plays a central role in powering a real-time emotional insight platform that helps parents understand how online gaming affects their child's emotional state. The platform relies on two structured datasets and three research studies to support gameplay monitoring, emotion prediction, and parent-facing visualizations.

2.1.1 Video Game Sales Dataset (DataCamp)

This dataset provides structured metadata on thousands of video games, including title, platform, release year, genre, and global sales. It offers a foundation for classifying the types of games children play and understanding how those game types correlate with emotional outcomes.

This data will inform:

- The categorization of game genres for emotion classification (e.g., FPS → Angry, Puzzle → Joy).
- Backend logic that tags game sessions with research-backed emotion types.

2.1.2 Online Gaming Behavior Dataset (Kaggle)

This dataset captures real or simulated gameplay behavior such as user age, gender, game ID, session duration, engagement levels, and behavior scores. It provides session-level visibility into how long games are played and how children engage with different types of games.

This data will inform:

- Real-time tracking of gaming sessions for each child.
- Emotion prediction models that adjust based on playtime and behavioral intensity.

2.1.3 Research Paper 1 – *Game Genre and Emotional Responses* (ScienceDirect)

This peer-reviewed study analyzes how various game genres impact players' emotional states, identifying specific emotional responses (e.g., anger, fear, joy) linked to different game types.

This research will inform:

- The mapping of game genres to primary emotional labels within the system.
- Personalized insights presented to parents based on the game type played by their child.

2.1.4 Research Paper 2 – *Emotional Impact of Gameplay Duration* (arXiv)

This open-access research explores the effect of increasing gameplay duration on emotional intensity, finding that longer sessions result in heightened emotional responses.

This research will inform:

- The scaling of emotion intensity based on playtime thresholds (e.g., >2 hrs = high intensity).
- The platform's alerting system for emotional risk after extended sessions.

2.1.5 Steam Dataset – *Game Metadata for Content Classification*

The Steam dataset complements existing emotion prediction logic by offering detailed game-level metadata. It is particularly useful for flagging games with violent or intense content using a rule-based classification model developed during data preparation.

This data informs:

- Tag-based violence classification (e.g., games tagged with "FPS", "Blood", or "War").
- Content warnings and emotion predictors based on Steam-defined metadata.
- Back-end logic that links genres and tags with age restrictions and behavioural triggers.

2.1.6 RAWG Video Games API – *Metadata Expansion & Machine Learning Input*

To enrich the emotion prediction model and enhance metadata completeness, the RAWG Video Games Database API was used to dynamically fetch game-level data (genre, tags, ESRB/ACB ratings, and user reviews).

- **Link**:https://rawg.io/
- https://www.classification.gov.au/

Use in Project:

- Supplemented missing or ambiguous metadata in the Steam and Sales datasets.
- Used to gather **ACB ratings** (Australian Classification Board) for region-specific content classification.
- Enabled training of a basic supervised learning model to classify emotional risk based on combined features: genre, tags, and age rating.

Machine Learning Use:

- A Random Forest Classifier was trained on a small subset of labeled data combining RAWG-fetched tags and ACB content ratings with emotion labels (from the rule-based research mappings).
- Input Features: Game genre, number of violent tags, age rating, average playtime.
- Output: Predicted emotion category (e.g., Anger, Calm, Excitement).
- Evaluation metrics: Accuracy, F1-score (on test split of the enriched dataset).

This ML approach complements the rule-based prediction model with a scalable alternative for future expansion.

2.1.7 User Input Logs

Users (parents) contribute session-level data via a form. This includes the child's mood, game details, and play duration. These logs:

- Complement model-generated emotion predictions
- Allow for real-world validation and dashboard personalization
- Form the basis for weekly summaries and emotional pattern tracking

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2.2 Data Preparation

2.2.1 Video Game Sales Dataset (DataCamp)

File Type: CSV
Libraries Used: pandas, numpy

Data Cleaning

- Load the dataset using pandas.read_csv().
- Use dropna() to remove rows with missing Genre, Platform, or Year_of_Release.
- Use str.lower().str.strip() and replace() to standardize genre names (e.g., "Shooter" → "FPS").

Data Wrangling

- Extract and rename key columns using pandas.rename() and column selection.
- Save the cleaned table as cleaned_games.csv using to_csv().

Outcome

- A clean game metadata table to support genre-based emotion prediction logic.

2.2.2 Online Gaming Behavior Dataset (Kaggle)

File Type: CSV
Libraries Used: pandas, numpy, datetime, re

Data Cleaning

- Load dataset with pandas.read_csv() and inspect using info() and isnull().
- Remove or impute missing values using dropna() or fillna().
- Normalize Engagement_Level with .str.lower().str.capitalize() and ensure PlayTime_Hours is in float format.

Data Wrangling

- Join with game metadata on Game_ID using merge() to assign Genre.
- Group and analyze sessions using groupby() for trend detection.
- Filter age-specific users (13–15) using boolean masks.

Outcome

- Structured, session-ready dataset with all values normalized and linked to game metadata.

2.2.3 Research Paper 1 – *Game Genre and Emotional Responses*

Manual Data Processing

Data Extraction

- Manually extract genre-to-emotion mappings from the study.
- Assign a baseline intensity score (e.g., FPS → Angry, intensity 8).

Data Formatting

- Create a table using pandas.DataFrame() and export to emotion_rules_genre.csv with to_csv().

Outcome

- Evidence-based genre-emotion table used in prediction logic.

2.2.4 Research Paper 2 – *Emotional Impact of Gameplay Duration*

Data Extraction

- Translate time-based emotion escalation patterns into playtime ranges.
- Define rules like:
  0–1.5 hrs = low, 1.5–3 hrs = medium, >3 hrs = high intensity.

Data Formatting

- Create a CSV or SQL-ready table with columns: min_playtime, max_playtime, intensity_multiplier.

Outcome

- Enhances emotion logic by scaling intensity based on session duration.

2.2.5 Steam Dataset – Data Cleaning and Preprocessing:

Libraries Used: pandas, re, datetime

Data Cleaning

- Parsed release_date into datetime format, discarding invalid entries.
- Converted multi-value fields (categories, genres, steamspy_tags) into Python lists.
- Extracted lower bounds from owners (e.g., "20,000–50,000" → 20000) for quantitative analysis.
- Handled missing or null values by imputation or safe drops.

Data Transformation – Violence Classification Logic

- Defined keyword-based violence levels using strong and mild indicators (e.g., "FPS", "Gore").
- Applied conditional logic:
  - +2 points: presence of strong violent tags (FPS, Gore, etc.)
  - +1 point: mild tags (Shooter, Battle, Combat)
  - +2 points: multiplayer + shooter combo
  - +2 points: age restriction ≥ 18

Final Classification Labels:

- **Highly Violent**: Score ≥ 5
- **Moderately Violent**: Score 3–4
- **Non-Violent**: Score ≤ 2

Outcome:

- Final cleaned dataset classified_steam_games.csv includes a new violence_level column.
- Enables real-time filtering and flagging of violent games on the parent dashboard.

2.2.6 RAWG API – Dynamic Data Integration

- **Libraries Used**: *requests, pandas, sklearn, json*

- **Steps**:
    - Queried game data using RAWG API endpoints with game title + platform.
    - Parsed JSON responses to extract: genres, tags, ESRB/ACB ratings, description, user_rating.
    - Normalized tag-based keywords using .lower().strip() for integration with the violence classification model.
    - Stored output in enriched_game_metadata.csv.

ACB Rating Processing:

- Extracted acb_rating from the API where available.
- Mapped common ACB categories (G, PG, M, MA15+, R18+) into a violence_severity_score scale.
- Integrated this score into the PostgreSQL Games table for use in visualizations and parent-facing warnings.

2.3 Data Storage

To support real-time emotion inference and parent-facing insights, this project adopts a structured, secure, and scalable approach to data storage. All cleaned datasets and rule-based logic are stored in a relational database using PostgreSQL, enabling efficient querying and integration with frontend dashboards

Storage Technology

- We use **PostgreSQL** as the primary relational database for structured data storage.
- Chosen for its support for advanced querying, joins, indexing, and ACID compliance.

Structured Data Handling

- Cleaned CSVs from the **Video Game Sales** and **Online Gaming Behavior** datasets are imported into dedicated tables (Games, Users, Sessions) using pgAdmin or SQL COPY command.
- Emotion logic extracted from research papers is manually converted into rule tables and stored as Emotion_Prediction.

Data Integration & Relationships

- Relationships are established using **foreign keys**:
  Sessions → Games via game_id,
  Sessions → Users via user_id,
  Emotions → Sessions via session_id.
- Emotion predictions are inferred using a **logical join** between Sessions, Games, and Emotion_Prediction.

User Input Logs Table

To support personalized emotional trend tracking, the platform collects and stores structured session data submitted by authenticated users (parents). This data includes:

- Mood after gameplay
- Game name and genre
- Date and duration of the session

These entries are stored in a dedicated table called `User_SessionLogs`, and each log is linked to the user via `user_id`. The table structure enables trend analysis, prediction validation, and personalized insights on the parent dashboard.

Schema Overview:

Column	Type	Description
`log_id`	UUID	Primary key
`user_id`	FK	Links to `Users` table
`mood`	TEXT	Self-reported emotion (e.g., Happy, Angry)
`game_name`	TEXT	Inputted game title
`game_genre`	TEXT	Inputted game genre
`play_date`	DATE	Date of session
`play_duration_minutes`	INT	Total computed duration
`timestamp_submitted`	TIMESTAMP	System timestamp of entry

This input enhances the emotional prediction logic by combining model-driven inference with real-world self-reported data, improving reliability and parent engagement.

Security & Access Control

- Database access is restricted to authenticated roles: admin (full access), analyst (read/query), and dashboard (read-only).
- All sensitive fields are pseudonymized (e.g., User_ID) and encrypted at rest using PostgreSQL's pgcrypto if needed.
- All write operations are logged for auditability.

Tools Used

- **PostgreSQL** for schema management and storage.
- **pgAdmin** for visual schema editing and manual inserts.
- **SQLAlchemy** (optional) for Python-based DB interactions.
- COPY, INSERT, or pandas.to_sql() for data ingestion.

Visualization & Retrieval

- Cleaned data is retrieved from PostgreSQL via API endpoints (Node.js) for integration with the parent dashboard.
- Aggregated data (e.g., weekly emotional trends) is preprocessed into materialized views for fast loading on the frontend.

The Steam dataset is stored as an independent table (Steam_Games) within PostgreSQL. It includes metadata such as genres, tags, required age, and the derived violence_level. Although it is not directly joined with gameplay data, it is used in standalone content analysis pipelines and supports dashboard references.

2.4 Data Archiving & Versioning

Effective data archiving and versioning are critical to maintaining the integrity, reproducibility, and traceability of the project. This section outlines how raw, processed, and cleaned datasets are organized, stored, and version-controlled.

2.4.1 Folder Structure and Organization

All datasets, scripts, and outputs are organized in a structured folder system:

/project-data

/raw → Original source datasets (downloaded from DataCamp, Kaggle, research papers)

/processed → Cleaned and preprocessed datasets ready for database import

/scripts → Python and SQL scripts for cleaning, transformation, and loading

/backup → Date-stamped backups of each dataset version

2.4.2 Archiving Tools

- **Google Drive** used as the primary archiving platform
- Each version is saved with a naming pattern like `cleaned_data_v1_2025-04-10.csv`
- GitHub (or a private repo) is used to track code and schema changes over time

2.4.3 Backup & Redundancy

- **Manual backups** currently stored locally and on Drive
- **Cloud-based backups** through Azure's built-in backup system for PostgreSQL (automated daily snapshots)
- Planned integration with Azure Blob Storage lifecycle rules for long-term retention

2.5 Database Design

The database supports core platform functions: tracking gaming sessions, predicting emotional responses, and presenting insights to parents. The schema is structured using a normalized relational model built in **PostgreSQL**.

2.5.1 Key Entities:

- **Users**
  Stores child demographics (user_id, age, gender).
- **Games**
  Game metadata including name, genre, platform, publisher.
- **Sessions**
  Logs gameplay activity (session_id, user_id, game_id, playtime_hours, engagement_level).

- **Emotions**
  Stores predicted outcomes (emotion_label, intensity_score, session_id, emotion_source).
- **Emotion_Prediction**
  Research-based rule table linking genre + playtime to emotion and intensity.

Relationships:

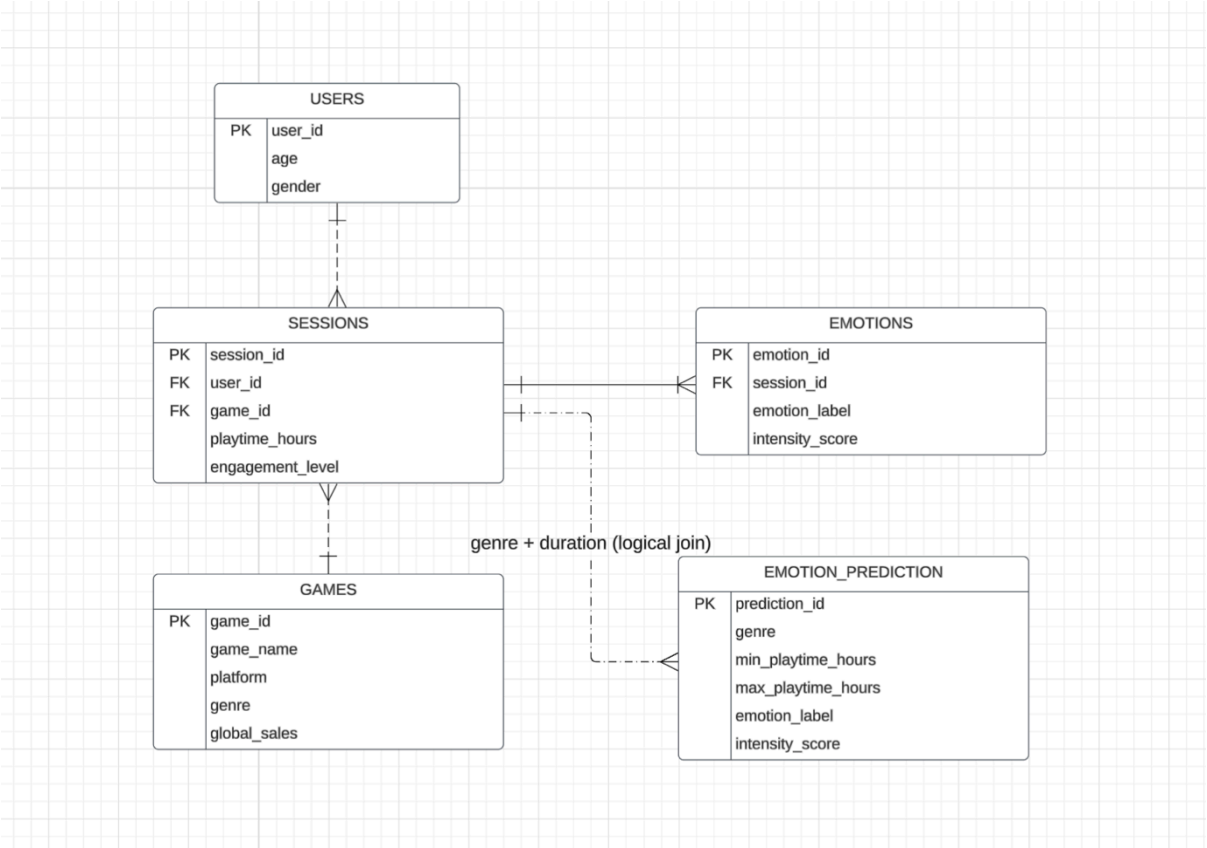
- Sessions.user_id → **Users**
- Sessions.game_id → **Games**
- Emotions.session_id → **Sessions**
- Emotion_Prediction is joined **logically** using genre + playtime_hours.

2.5.2 Tools & Implementation:

- Database: **PostgreSQL**
- Diagramming: **Lucidchart**.
- Queries: Handled via SQL
- Tables built using “CREATE TABLE” with foreign keys and indexing for performance.

Design Highlights:

- Normalized for data integrity and scalability
- Supports real-time emotion prediction
- Enables efficient visualizations (e.g., playtime vs emotion graphs)
- Easily extendable for future features like parental alerts or weekly summaries



2.6 Data Analytics and Modelling Techniques

Analytics is used to drive emotional predictions, personalize parental insights, and visualize gameplay impact in real-time. Each dataset and research-based rule contributes to interpreting player behaviour and forecasting emotional outcomes.

2.6.1 Hindsight Insights (Historical Patterns)

- Based on historical session logs, the system identifies frequent emotional triggers (e.g., FPS games causing anger after 2+ hrs).
- Game genre and playtime combinations are analyzed to generate weekly summaries and insights.

- Steam metadata is used independently to classify games by violence level. We can analyse the number of violent games.

Use in Platform:

- “Your child felt angry in 4 of 6 sessions this week.”
- Dashboard visualizations of emotion trends over time.

2.6.2 Foresight Insights (Predictive Feedback)

- Emotion is predicted in real-time using genre + playtime rules.

- If thresholds are exceeded, the system anticipates emotional escalation (e.g., frustration, fatigue).

Use in Platform:

- "Warning: This session may trigger frustration."
- Real-time recommendations such as breaks or game switches.

2.6.3 Visual Insights (Dashboard & Charts)

- Visualizations show emotion intensity vs playtime (line graphs, emoji scales).
- Parents receive summaries like "Most triggered emotion: Angry," and tips based on behavior trends.

Use in Platform:

- Line chart of playtime vs emotional intensity.
- Emoji-based feedback + action suggestions.

2.6.4 Critical Thinking Behind Modeling Choices

The modeling choices in this project were made deliberately to balance accuracy, interpretability, and ethical responsibility, particularly because the system is designed for a non-technical audience (parents) and involves sensitive insights about children's emotional states.

Simplicity and Interpretability : Given the project's goal of delivering clear, relatable insights to parents, we prioritized **rule-based models** over black-box machine learning algorithms. For example:

- We used **genre-to-emotion mappings** and **playtime thresholds** extracted from peer-reviewed research instead of opaque predictive models.

- This approach ensures that parents can understand the connection between a child's gameplay and the predicted emotional outcome.

Evidence-Based Foundations : Rather than relying on subjective assumptions or unvalidated algorithms, we grounded our emotion predictions in:

- Research Paper 1 → genre-emotion relationships.

- Research Paper 2 → duration-intensity escalation patterns.

- Steam metadata → violence and age-appropriateness classification.

This gives our system both scientific credibility and practical relevance.

Scenario-First Thinking : The analytics pipeline was designed with real-world scenarios in mind:

- Instead of abstract metrics, the system delivers insights such as:
    - “Your child played high-violence games for 3+ hours → emotional risk: frustration or anger.”
- This approach improves relevance and supports constructive conversations between parents and children.

Progressive Complexity : We intentionally structured the system to **start simple** (rule-based models) while preparing for:

- Future machine learning integration.
- Advanced predictive models trained on real-world behavior (once ethically sourced datasets become available).
- Bias testing and fairness evaluations to ensure inclusive and ethical AI.

User-Centric Communication : Finally, all insights are communicated through:

- Visual dashboards (e.g., emoji scales, line graphs).
- Simple recommendations (“Suggest a break after 2 hours of play”).
- Transparent explanations, avoiding technical jargon or overwhelming parents.

While the rule-based system offers transparency, a prototype ML classifier was developed using enriched metadata from the RAWG API. This model served as a feasibility test for future integration of intelligent, scalable predictions. ACB ratings specifically offered culturally relevant filtering aligned with Australian parental concerns.

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3. Ethical, Legal, and Privacy Considerations

The use of behavioral, gaming, and emotional data—especially in contexts involving minors—raises critical ethical, legal, and privacy considerations. This section outlines the safeguards, principles, and responsibilities adopted to ensure data is collected, processed, and stored securely and respectfully.

3.1 Privacy and Data Handling

- All external datasets (Video Game Sales, Online Gaming Behavior, Steam, and Research Papers) are **open-access and de-identified**, ensuring no personally identifiable information (PII) is collected from those sources.

- **User authentication** is handled via **Firebase Authentication**, supporting both email/password login and Google OAuth sign-in.
    - Passwords are hashed and securely managed by Firebase; the application **never stores plaintext credentials**.
    - Only **session-based login persistence** is enabled (`browserSessionPersistence`) to reduce the risk of unauthorized access on shared devices.
- After login, users may voluntarily submit gameplay session data through a structured modal form, including:
    - Mood/emotion after gameplay
    - Game name and genre
    - Date of gameplay
    - Duration (hours and minutes)
- This **user-generated data is stored in the `User_SessionLogs` table**, pseudonymized using either the Firebase UID or a hashed `user_id`.
    - No personal names, free-text inputs, or sensitive psychological data are collected.
    - Mood data is stored as categorical values (e.g., "Happy", "Irritable") for trend analysis only.
- The entire PostgreSQL database (hosted on Azure) implements **encryption at rest**. Access is strictly **role-based**:
    - Admin → Full access
    - Analyst → Read/query access only
    - Dashboard → Read-only for frontend visualizations
- All data submissions occur over **HTTPS-secured API calls**. The frontend includes client-side validation to prevent malformed or incomplete entries.
- Dataset backups are versioned and archived using structured folder hierarchies, Google Drive, and Azure snapshot tools.

3.2 Legal Considerations

- All third-party datasets are governed by open-use licenses:
    - **Creative Commons BY 4.0**: Video Game Sales, Online Gaming Behavior, Steam Dataset
    - **Licensed or Open-Access**: Research papers via ScienceDirect (educational license) and arXiv (open access)
- The platform collects **no direct data from minors**. All session logs are entered by parents voluntarily, based on observation, ensuring:
    - Compliance with **COPPA** (US), **GDPR for Minors** (EU), and the **Australian Privacy Act 1988 (Cth)**.
    - Protection from legal liabilities involving child data or biometric/emotional surveillance.

- The entire application is used in a **non-commercial, academic setting** as part of a university research project.
- All licensing terms are respected across data ingestion, transformation, modeling, and visualization.

3.3 Ethical Use of AI & Algorithmic Bias

- The platform currently uses **rule-based models** for emotion prediction, based on:
  - Peer-reviewed research on **game genre-emotion mappings**
  - Studies of **duration-emotion escalation patterns**
- A **Random Forest prototype** was developed using enriched metadata (e.g., RAWG API), but is not deployed in production. This prototype served only to test future scalability.
- Planned ethical AI practices include:
  - **Bias testing** on gender, age, and gameplay preferences
  - Implementation of **Explainable AI (XAI)** to help parents understand prediction logic
  - **Fairness audits** to ensure that model performance is inclusive and balanced
- Predicted emotions are presented as **guidance, not diagnosis**, maintaining user trust and reducing risk of misinterpretation.

3.4 Ethical Intent of the Project

- The platform is designed to **support, not surveil**. It empowers parents with insights that encourage healthy discussions and promote emotional awareness in digital gaming contexts.

- All data input is **voluntary, structured, and limited to gameplay context** — there is no passive monitoring, screen tracking, or third-party surveillance.

- The system avoids complex, technical outputs. Instead, it focuses on **relatable language and visual cues** such as emoji scales and trend charts.

- Scenarios and insights are crafted to be **age-appropriate and respectful**, with an emphasis on building digital resilience and empathy through reflection.

4. Understanding the Problem Domain & Future Challenges

4.1 Problem Domain: Online Gaming, Emotional Wellbeing, and Adolescent Behavior

Adolescents aged 13–15 are at a critical developmental stage, where emotional regulation, empathy, and social interaction are rapidly evolving. Online gaming—while offering entertainment and social connection—also introduces risks such as aggression, frustration, and emotional dysregulation, particularly in genres like first-person shooters (FPS) and multiplayer battle games.

Research has shown that prolonged gameplay and certain game types can heighten emotional responses, including anger, sadness, and joy, often without players' or parents' full awareness. Many parents lack the digital literacy or tools to recognize when gaming crosses from healthy engagement into emotional risk.

This platform is designed to bridge that gap by providing parents with timely, evidence-based insights into their child's gaming behavior, emotional patterns, and potential risks, using real-world data and behavioral models.

4.2 User Needs Addressed

User Group	Needs
Parents	Tools and insights to understand their child's emotional state during gaming; early warning of frustration or fatigue; tips for supportive intervention.

The platform uses accessible visualizations, relatable language, and actionable recommendations to meet these needs across stakeholder groups.

4.3 Future Integration & Challenges

4.3.1 Planned Dataset Additions

- **Real-world emotion datasets:** To improve accuracy of emotion predictions.
- **Platform-specific behavior logs (e.g., Xbox, PlayStation, Steam):** To expand coverage across devices.

4.3.2 Anticipated Challenges and Strategies

Challenge	Strategy
Dataset Bias	Validate models with diverse datasets; monitor for overfitting to narrow player profiles.
Evolving Game Trends and Slang	Update genre mappings and violence classifiers regularly; apply dynamic NLP models to capture emerging slang.
Explainability and Trust	Provide transparent explanations of emotion predictions; involve parents and teens in co-designing feedback mechanisms.